

ANEESH KOMANDURI

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EDUCATION

University of Arkansas Fayetteville, Arkansas
Doctor of Philosophy (Ph.D.), Computer Science Aug 2021 - May 2026

- **Dissertation** - Toward Causal Generative Modeling: From Representation Learning to Controllable Generation

Master of Science (M.S.), Computer Science 2021 - 2024
Bachelor of Science (B.S.), Computer Science & Mathematics 2017 - 2021

SKILLS

- **Programming Languages:** Python, C/C++, Java, Javascript, SQL
- **Frameworks:** PyTorch, Tensorflow, scikit-learn, Pyro, Transformers, Deepspeed, Accelerate
- **ML Concepts:** Large (Vision) Language Models (LLM/LVLM), Multimodal Models, Diffusion Models, Parameter-efficient Fine-tuning (e.g., LoRA), LLM Agents, Retrieval Augmented Generation (RAG), Variational Autoencoders, Causal Machine Learning, Graph Neural Networks, Machine Unlearning, Multi-GPU Distributed Computing
- **Technologies:** LangChain, MCP, A2A, Flask, Django, AWS, Spark, Databricks, Postgres

EXPERIENCE

University of Colorado Anschutz School of Medicine Aurora, Colorado
Postdoctoral Research Scientist June 2026 - Present

- Researching foundational methods in natural language processing for clinical healthcare applications in the *Language, Reasoning, and Knowledge (LARK)* lab, Department of Biomedical Informatics
- Working on developing robust and interpretable NLP methods for clinical decision making, including uncertainty quantification, reasoning in large language models, and multi-agent systems

University of Arkansas Fayetteville, Arkansas
Graduate Research Assistant Oct 2021 - June 2026

Trustworthy Vision-Language Models

- Engineered a robust **benchmarking framework** for **causal reasoning in LVLMs**; annotated three physical system image datasets, formulated three novel causal tasks, and benchmarked 9 models under **zero-shot**, **few-shot**, and **chain-of-thought prompting** [EMNLP'25]
- Established a realistic setting for **unlearning sensitive information in LVLMs** and designed a **resource-efficient, cross-attention guided** method that targets least-attended visual tokens, significantly improving unlearning-utility trade-offs compared to full fine-tuning baselines [LockLLM@NeurIPS'25]

Causal Generative Modeling

- Proposed and implemented a theoretically novel and empirically robust **VAE framework** for **causal representation learning** on high-dimensional data, outperforming state-of-the-art baselines by 15%–20% in disentanglement metrics [IJCAI'24]
- Designed a novel interpretable **causally-aware diffusion model** for generating high-quality **image counterfactuals** that outperforms state-of-the-art baseline methods in the quality of the generated counterfactuals along both quantitative and qualitative metrics [ECAI'24]
- Synthesized theoretical and empirical **causal generative modeling** research in a 70-page technical survey, defining key methods for **causal representation learning** and **counterfactual generation**, standardizing evaluation metrics, and identifying future research directions [TMLR'24]

INTERNSHIP EXPERIENCE	NEC Laboratories America, Inc. Princeton, New Jersey <i>Research Scientist Intern</i> May 2025 - Aug 2025 • Developed core components of an in-house LLM-based AI chatbot system for multi-aspect IT Ticket service requests data • Designed a hybrid concept and embedding refinement framework for historical ticket data to achieve efficient retrieval augmented generation (RAG) for new user requests • Implemented a weakly-supervised contrastive learning algorithm to learn context-rich refined semantic embeddings that improved RAG performance by 20% compared to baselines
RELEVANT PUBLICATIONS	<u>Aneesh Komanduri</u> and Xintao Wu. Leveraging Foundation Models for Causal Generative Modeling. <i>Under Review at IEEE International Conference on Data Mining.</i> 2026. Karuna Bhaila, <u>Aneesh Komanduri</u>, Minh-Hao Van, and Xintao Wu. Cross-Modal Attention Guided Unlearning in Large Vision-Language Models. <i>Lock-LLM Workshop@NeurIPS.</i> 2025. <i>Under Review at 2026 Conference on Language Modeling.</i> <u>Aneesh Komanduri</u>, Karuna Bhaila, and Xintao Wu. CausalVLBench: Benchmarking Visual Causal Reasoning in Large Vision-Language Models. <i>Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP Main).</i> 2025. <u>Aneesh Komanduri</u>. Toward Causal Generative Modeling: From Representation to Generation. <i>Proceedings of AAAI Conference on Artificial Intelligence (AAAI).</i> 2025. <u>Aneesh Komanduri</u>, Chen Zhao, Feng Chen, and Xintao Wu. Causal Diffusion Autoencoders: Toward Counterfactual Generation via Diffusion Probabilistic Models. <i>Proceedings of 27th European Conference on Artificial Intelligence (ECAI).</i> 2024. *Also appeared in non-archival <i>Generative Models for Computer Vision Workshop at CVPR 2024</i> <u>Aneesh Komanduri</u>, Yongkai Wu, Feng Chen, and Xintao Wu. Learning Causally Disentangled Representations via the Principle of Independent Causal Mechanisms. <i>Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI),</i> 2024. *Also appeared in non-archival <i>Causal Representation Learning Workshop at NeurIPS 2023</i> <u>Aneesh Komanduri</u>, Xintao Wu, Yongkai Wu, and Feng Chen. From Identifiable Causal Representations to Controllable Counterfactual Generation: A Survey on Causal Generative Modeling. <i>Transactions on Machine Learning Research (TMLR).</i> 2024.
HONORS AND AWARDS	Silver Reviewer Award May. 2026 <i>Awarded by International Conference on Machine Learning (ICML)</i> 1st Place Graduate Student Poster @ NSF 2024 DART Conference (\$1,500) Sep. 2024 <i>Awarded by Arkansas Economic Development Commission (AEDC)</i> Doctoral Academy Fellowship (\$48,000) 2021-2025 <i>Awarded by University of Arkansas Graduate School and International Education</i>
SERVICE	Conference Reviewer: ICML'26, AAAI'26, ECAI'25, KDD'25, PAKDD'25, IJCAI'25, ICML'25, ICLR'25, NeurIPS'24, LoG'24, ICMLA'24 Journal Reviewer: TMLR, DMLR, IJDSA, Pattern Recognition, IEEE Access, IEEE TBD Workshop Reviewer: CaLM@NeurIPS'24, SPIGM@ICML'24